

A Stage-Wise Evaluation of YOLOv11s for PCB Surface Defect Detection: Input, Backbone, Neck, Head, and Inference-Level Modifications

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Abstract—Automated optical inspection of printed circuit boards (PCBs) increasingly relies on deep object detectors, and many recent studies improve performance by adding attention modules, extra detection heads, or multi-scale feature fusion. This work evaluates whether such modifications improve a strong YOLOv11s (You Only Look Once version 11 small) baseline for PCB surface defect detection on the DsPCBSD+ (Dataset of Printed Circuit Board Surface Defects) dataset. Instead of treating experiments as unrelated trials, the study is organized along the YOLO (You Only Look Once) detection pipeline: input/data processing, backbone feature extraction, neck feature fusion, detection-head prediction, and inference/post-processing. The tested variants include heavy augmentation, rare-class oversampling, Coordinate Attention, CBAM (Convolutional Block Attention Module), a P2 (Pyramid Level 2) small-object head, BiFPN-style (Bidirectional Feature Pyramid Network) weighted fusion, EUCB (Efficient Upsampling Convolution Block) upsampling, EdgeMix augmentation, high-resolution training, SAHI (Slicing Aided Hyper Inference), and test-time augmentation. The main finding is that none of the tested modifications surpasses the plain YOLOv11s baseline, which achieves mAP@0.5 (mean Average Precision at an Intersection over Union threshold of 0.5) = 0.839 and mAP@0.5:0.95 (mean Average Precision across Intersection over Union thresholds from 0.5 to 0.95) = 0.499. Some input- and neck-level changes remain competitive and improve specific weak classes, but the gains are not consistent across all defects. The results indicate that the remaining errors are less related to model capacity and more related to weak-class ambiguity, low contrast, and fine defect texture. We additionally report the baseline’s parameter and compute footprint to frame the findings for edge deployment.

Index Terms—PCB defect detection, YOLOv11, object detection, attention mechanisms, feature fusion, automated optical inspection.

I. INTRODUCTION

Printed circuit boards (PCBs) are the foundation of most electronic products, and surface defects introduced during fabrication can cause functional failures if they reach assembly. Defects such as shorts, opens, spurs, mouse bites, conductor scratches, and foreign objects must therefore be detected reliably during automated optical inspection (AOI). Deep object detectors, especially YOLO-family models, are

attractive for this task because they combine localization, classification, and real-time inference.

Recent PCB defect detection studies often improve YOLO-style detectors by adding architectural modules such as attention blocks, small-object detection heads, multi-scale fusion layers, or edge-aware refinements [1]–[4]. These methods are usually motivated by the small and subtle nature of PCB defects. However, it is not always clear whether such improvements remain beneficial once the baseline detector is already strong and well trained.

This work asks a practical question: *where should YOLOv11s be improved for PCB defect detection?* To answer this, we evaluate modifications at different stages of the YOLO pipeline. Input-level experiments test whether weak visual cues can be improved before the image enters the network. Backbone-level experiments test whether attention improves feature extraction. Neck-level experiments test whether feature fusion and upsampling improve multi-scale representation. Detection-head experiments test whether a high-resolution P2 head improves small-defect localization. Finally, inference-level experiments test whether slicing or test-time augmentation can recover missed detections without retraining.

The contributions of this work are:

- A strong YOLOv11s baseline for DsPCBSD+ PCB surface-defect detection.
- A stage-wise evaluation of modifications at the input/data, backbone, neck, detection-head, and inference levels.
- A controlled comparison of attention modules, oversampling, P2 detection, BiFPN-style fusion, EUCB upsampling, EdgeMix augmentation, high-resolution training, SAHI, and test-time augmentation.
- A negative but useful finding: none of the tested modifications consistently surpasses the plain YOLOv11s baseline, supported by an analysis of the baseline’s parameter and compute footprint.
- An error-based interpretation showing that the hardest classes—especially conductor scratch and conductor for-

eign object—are limited by visual ambiguity and low-contrast texture rather than prediction resolution alone.

II. RELATED WORK

PCB defect detection. YOLO-based detectors are widely used for PCB inspection because they provide a good speed–accuracy trade-off. Prior work has explored small-object heads, lightweight network designs, multi-scale feature extraction, and edge-aware detection for PCB surface defects [1]–[4]. These methods motivate the stage-level modifications evaluated in this work.

Attention mechanisms. Attention modules are often added to convolutional networks to emphasize informative channels or spatial regions. Squeeze-and-Excitation [5] performs channel reweighting, while CBAM [6] applies both channel and spatial attention; Coordinate Attention further preserves positional information during attention. In this work, attention is tested at the backbone level to determine whether improved feature selection helps distinguish subtle defects from normal PCB texture.

Multi-scale fusion and small-object detection. Feature Pyramid Networks [7] and BiFPN-style fusion [8] combine information across multiple feature scales. Such fusion is relevant to PCB defects because thin scratches and small foreign objects may require both shallow spatial detail and deeper semantic context. Similarly, a P2 detection head adds a higher-resolution prediction map, which may help small objects but may also increase false positives.

Inference-time enhancement. SAHI [9] applies detection on overlapping image slices to improve small-object visibility, while test-time augmentation and Weighted Boxes Fusion [10] combine predictions from transformed versions of the same image. These methods do not change model weights and are evaluated as inference-level modifications.

Backbone. We build on YOLOv11 [11], whose architectural refinements over prior YOLO generations [12] provide the strong starting point under which we test the value of further additions.

III. DATASET

We use the DsPCBSD+ dataset [13], a public PCB surface-defect detection dataset published in *Scientific Data*. It defines nine defect categories, shown in Table I. The classes vary significantly in visual appearance and difficulty: hole breakout is often clear and circular, while conductor scratch and conductor foreign object are thin, irregular, low contrast, and easily confused with copper traces or background PCB texture. All experiments use the released train, validation, and test splits.

IV. METHODOLOGY

This work follows a stage-wise experimental design based on the YOLO detection pipeline. Each experiment is assigned to the pipeline stage it is intended to improve: input/data processing, backbone feature extraction, neck feature fusion, detection-head prediction, and inference/post-processing. This

TABLE I
DsPCBSD+ DEFECT CLASSES (INDEX ORDER).

ID	Abbrev.	Defect name
0	SH	Short
1	SP	Spur
2	SC	Spurious copper
3	OP	Open
4	MB	Mouse bite
5	HB	Hole breakout
6	CS	Conductor scratch
7	CFO	Conductor foreign object
8	BMFO	Base material foreign object

allows each modification to test a specific hypothesis about the source of detection errors.

All models are based on YOLOv11s [11] and trained on a single GTX 1650 Ti GPU (4 GB VRAM), which constrains batch size and motivates mixed-precision training. The baseline is a vanilla YOLOv11s detector trained on the original DsPCBSD+ images at 640 px with the default backbone, neck, and P3–P5 detection heads. Precision, recall, mAP@0.5, and mAP@0.5:0.95 are used as the main evaluation metrics.

A. Input and Data-Level Experiments

Input-level experiments test whether detection is limited by image representation rather than architecture. **Heavy augmentation** tests whether stronger photometric and geometric variation improves generalization. **Rare-class oversampling** increases the frequency of weak or under-performing classes during training.

EdgeMix is implemented as an offline training-time augmentation. For each training image, the original image is kept, and an additional edge-enhanced copy is generated. The enhancement is applied in the LAB color space: the image is converted from RGB/BGR to LAB, CLAHE is applied to the luminance channel to improve local contrast, and a mild unsharp mask is then applied to emphasize fine boundaries and texture details. The enhanced luminance channel is merged back with the original color channels and converted back to RGB/BGR. The corresponding label file is copied unchanged because the geometric position of each defect is not modified. Thus, EdgeMix doubles the training images by mixing original samples with enhanced copies, while validation and test images remain unchanged.

Fig. 1 shows the difference between the original PCB image and the filtered EdgeMix version. The purpose is to make subtle defects such as conductor scratches and foreign objects more visible during training without changing the validation or test image distribution. **High-resolution training** increases the input size from 640 px to 960 px to test whether finer spatial detail improves localization.

B. Backbone-Level Experiments

Backbone-level experiments test whether improved feature extraction helps the detector distinguish subtle defects from PCB background texture. **Coordinate Attention** is inserted

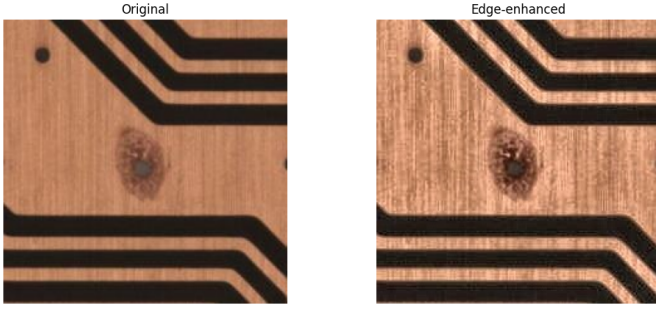


Fig. 1. Comparison between the original PCB image and the edge-enhanced image used in EdgeMix augmentation. The filtered version emphasizes local contrast, boundaries, and fine texture cues while preserving the same bounding-box annotations.

into the backbone to preserve positional information while reweighting channels. **CBAM** [6] is also tested because it combines channel and spatial attention. A **CBAM + over-sampling** configuration evaluates whether attention becomes more effective when weak classes are shown more frequently. A **CBAM fine-tuning** variant inserts attention into an already-trained checkpoint to test attention transfer.

C. Neck-Level Experiments

The YOLO neck combines shallow spatial features with deeper semantic features. A **BiFPN-style weighted fusion** neck [8] replaces standard concatenation with learnable weighted fusion, allowing the model to learn how much each feature scale should contribute. **EUCB refinement** is tested as a lightweight learnable upsampling module: the Efficient Up-Convolution Block introduced in the EMCAD decoder [14], which replaces a standard up-sampling stage with a depth-wise up-sampling unit ($\times 2$ nearest-neighbour up-sample, a depth-wise 3×3 convolution with batch normalisation and activation, then a 1×1 channel projection). These experiments test whether better feature recovery and fusion help thin or low-contrast defects.

D. Detection-Head-Level Experiments

Detection-head experiments test whether the default YOLOv11s prediction scales are sufficient. A **P2 small-object head** adds an additional stride-4 high-resolution prediction layer, producing detections at P2, P3, P4, and P5 instead of only P3–P5. A combined **P2 + BiFPN** configuration tests whether weighted feature fusion can reduce the additional noise introduced by the P2 head.

E. Inference and Post-Processing Experiments

Inference-level experiments do not modify trained weights. **Test-time augmentation (TTA)** combines predictions from transformed views of the same image. **SAHI** sliced inference [9] applies the detector to overlapping image slices to test whether small defects can be recovered by increasing their apparent scale during inference.

F. Training Protocol

The experiments are divided into primary training runs and fine-tuning runs. The baseline, heavy augmentation, Coordinate Attention, CBAM, rare-class oversampling, and P2 experiments are treated as primary, from-scratch runs. Later experiments—EUCB, BiFPN-style fusion, EdgeMix, high-resolution training, and P2 + BiFPN—are fine-tuned from a shared extended-baseline (100-epoch) checkpoint using a lower learning rate (5×10^{-5}). On a single 4 GB GPU, full from-scratch runs for every architecture are prohibitively expensive, and fine-tuning from a common checkpoint isolates the marginal effect of each module at fixed compute. This introduces a confound: fine-tuned variants inherit the representation quality of the parent checkpoint and are not perfectly equivalent to from-scratch comparisons. These runs are therefore interpreted as tests of whether a modification can recover or improve performance when added to an already-trained YOLOv11s pipeline, not as universal evidence that the module can never help.

V. EXPERIMENTAL RESULTS

A. Evaluation Protocol

All models are evaluated using precision, recall, mAP@0.5, and mAP@0.5:0.95 at the best validation epoch. SAHI is reported separately as an inference-only test. The goal is not only to compare total mAP, but also to determine whether any stage-level modification improves the weak classes.

B. Overall Stage-Wise Comparison

Table II summarizes the experiments according to the YOLO stage targeted by each modification, and Fig. 2 places them side by side. The vanilla YOLOv11s baseline remains the strongest overall detector, reaching mAP@0.5 = 0.839 and mAP@0.5:0.95 = 0.499. None of the tested input-, backbone-, neck-, detection-head-, or inference-level modifications surpasses this reference.

The most competitive non-baseline results come from input-level and neck-level changes. EdgeMix and high-resolution training remain close to the baseline, suggesting that preserving local visual detail is useful. EUCB and BiFPN-style fusion also remain competitive but do not improve the overall result. In contrast, P2-based detection-head modifications produce the largest degradation, indicating that increasing high-resolution prediction density adds more noise than useful signal for this dataset.

C. Training Dynamics

Fig. 3 shows the baseline training curves. Validation performance rises quickly and reaches its best result at the end of the 50-epoch run. Extending the baseline to 100 epochs did not improve the result, reaching mAP@0.5 = 0.837 and mAP@0.5:0.95 = 0.496. This indicates that the baseline was not simply under-trained; the model had already reached a strong operating point, making later architectural gains difficult.

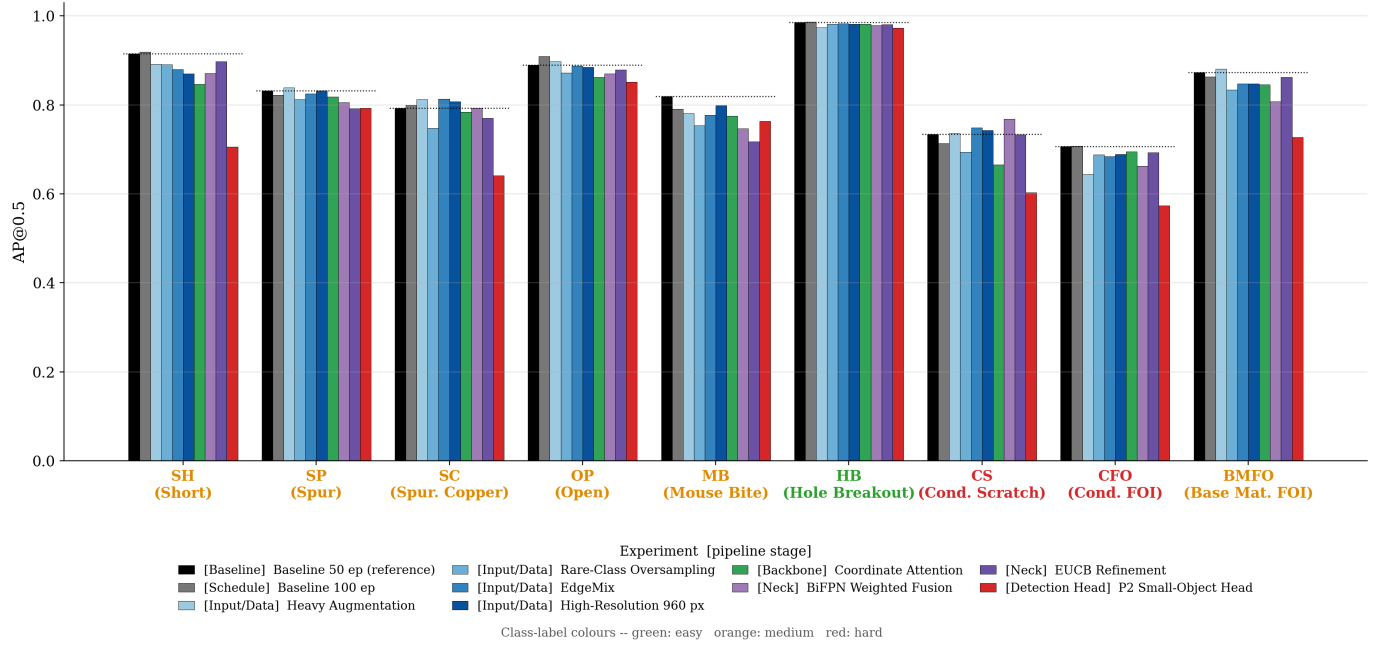


Fig. 2. Per-class detection accuracy (AP@0.5) across all experiments, with bars coloured by the YOLO pipeline stage each modification targets and the dotted line marking the Baseline (50 ep) per class. No variant exceeds the baseline on any class; hole breakout (HB) is easy for all models, while conductor scratch (CS) and conductor foreign object (CFO) remain the hardest.

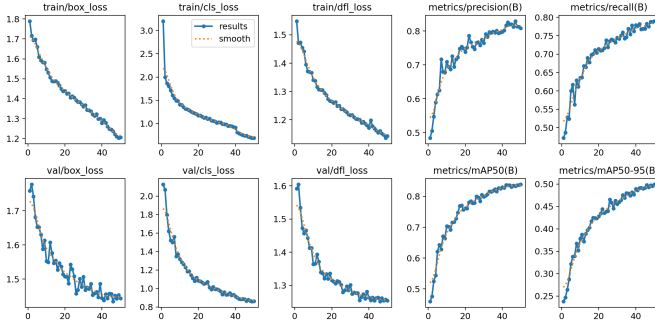


Fig. 3. Baseline YOLOv11s training dynamics. Validation metrics rise smoothly, and the 50-epoch baseline defines the strongest overall result used for comparison.

D. Input and Data-Level Findings

Input-level changes produced the strongest non-baseline results. EdgeMix reached $\text{mAP}@0.5 = 0.827$ and $\text{mAP}@0.5:0.95 = 0.488$, while high-resolution training at 960 px achieved $\text{mAP}@0.5 = 0.828$ and $\text{mAP}@0.5:0.95 = 0.487$. These results suggest that preserving local visual detail is more useful than adding complexity to the detection head. However, neither surpassed the baseline, meaning that simply enhancing details or increasing resolution was not sufficient to solve the hardest classes. Rare-class oversampling also did not help: although designed to improve weak-class exposure, it reduced $\text{mAP}@0.5:0.95$ to 0.451, suggesting the weak-class problem is not only a class-frequency issue—repeating ambiguous examples can overemphasize them

without improving generalization.

E. Backbone-Level Findings

Backbone attention did not improve the detector. Coordinate Attention was the strongest attention-only model ($\text{mAP}@0.5 = 0.813$, $\text{mAP}@0.5:0.95 = 0.454$) but remained well below the baseline; CBAM performed worse, and the CBAM fine-tuning run collapsed with very low recall. These results suggest the main limitation is not insufficient feature reweighting in the backbone: attention can emphasize features that are already present, but it cannot reliably resolve weak classes whose appearance is ambiguous or close to the PCB background.

F. Neck-Level Findings

Neck-level changes were more stable than detection-head changes. EUCB reached $\text{mAP}@0.5 = 0.814$ ($\text{mAP}@0.5:0.95 = 0.472$) and BiFPN-style weighted fusion reached $\text{mAP}@0.5 = 0.811$ ($\text{mAP}@0.5:0.95 = 0.476$), close to each other and below the baseline. The class-wise BiFPN result is informative: it achieved conductor scratch (CS) $\text{mAP}@0.5 = 0.768$ but conductor foreign object (CFO) remained lower at $\text{mAP}@0.5 = 0.663$. Better feature fusion can help thin scratch-like defects, but it does not fully resolve visually ambiguous foreign-object defects.

G. Detection-Head-Level Findings

The P2 detection-head experiments were the weakest stage-level results. The P2 model reached only $\text{mAP}@0.5 = 0.736$ ($\text{mAP}@0.5:0.95 = 0.403$); adding BiFPN-style fusion improved stability to $\text{mAP}@0.5 = 0.778$ ($\text{mAP}@0.5:0.95 = 0.432$) but remained far below the baseline. The interpretation

TABLE II
STAGE-WISE COMPARISON OF YOLOv11s MODIFICATIONS. METRICS ARE BEST-EPOCH VALIDATION RESULTS UNLESS STATED OTHERWISE.

Stage	Experiment	Epoch	P	R	mAP@0.5 / mAP@0.5:0.95	Main finding
Baseline	YOLOv11s 640	50/50	0.809	0.790	0.839 / 0.499	Strongest overall model; reference for all later experiments
Input/Data	Heavy Augmentation	48/50	0.790	0.778	0.830 / 0.488	Stronger augmentation did not improve the baseline
Input/Data	Rare-Class Oversampling	18/28	0.810	0.779	0.810 / 0.451	Oversampling reduced localization quality
Input/Data	EdgeMix Augmentation	best	0.823	0.766	0.827 / 0.488	Competitive input-level result; below baseline
Input/Data	High-Resolution 960 px	best	0.824	0.770	0.828 / 0.487	Competitive with EdgeMix; did not exceed baseline
Backbone	Coordinate Attention	49/50	0.788	0.767	0.813 / 0.454	Best attention-only run, but below baseline
Backbone	CBAM	49/50	0.758	0.679	0.769 / 0.430	Full backbone attention degraded performance
Backbone+Data	CBAM + Oversampling	best	0.75	0.72	0.761–0.765 / 0.419–0.420	Oversampling did not make CBAM competitive
Backbone	CBAM Fine-Tuning	28/75	0.932	0.047	0.043 / 0.016	Transfer/fine-tuning collapse with very low recall
Neck	EUCB Refinement	30/30	0.800	0.773	0.814 / 0.472	Valid feature-recovery test; competitive but below baseline
Neck	BiFPN-Style Weighted Fusion	30/30	0.795	0.769	0.811 / 0.476	Helped CS more than CFO; no overall gain
Detection Head	P2 Small-Object Head	24/39	0.725	0.762	0.736 / 0.403	Extra high-resolution head introduced noise
Head + Neck	P2 + BiFPN	stopped	0.775	0.737	0.778 / 0.432	Weighted fusion improved P2 stability, but far below baseline
Inference	Extended Baseline + TTA	–	–	–	0.821 / 0.495	Best inference-time setting, but still below baseline
Inference	SAHI	test	0.603	0.594	0.531 / 0.305	Slicing alone was unsuitable without slice-based training

is a scale mismatch. Quantitatively, over the DsPCBSD+ training labels the mean ground-truth defect box measures 110×102 px (about 17% of the 640 px image side), the median defect spans 79 px, and only 8% of instances fall below the 32 px COCO “small” threshold while 44% exceed 96 px (Table III). PCB defects here are predominantly mid-scale patches rather than the sub-16 px objects a stride-4 P2 branch targets, so the extra head increased sensitivity to small PCB texture and false positives instead of solving the weak-class problem.

H. Inference-Level Findings

Inference-level methods did not recover the missed weak-class detections. TTA achieved $\text{mAP@0.5} = 0.821$ ($\text{mAP@0.5:0.95} = 0.495$), close to but not exceeding the baseline. SAHI performed poorly ($P = 0.603$, $R = 0.594$, $\text{mAP@0.5} = 0.531$, $\text{mAP@0.5:0.95} = 0.305$): since the detector was trained on full images, slicing at inference time changed the image context and produced unstable detections, so slicing alone was unsuitable without slice-based training.

I. Weak-Class Error Analysis

Fig. 4 shows the normalized confusion matrix of the best-performing baseline model. The error structure is consistent

with the visual nature of the defects: hole breakout (HB) is detected with very high accuracy because of its clear circular structure ($\text{AP@0.5} \approx 0.99$), while conductor scratch (CS, $\text{AP@0.5} \approx 0.73$) and conductor foreign object (CFO, $\text{AP@0.5} \approx 0.71$) remain the hardest classes—thin, irregular, low contrast, and easily confused with copper traces or background. The stage-wise results show that different modifications help different parts of the problem, but none solves both weak classes: BiFPN fusion improves CS more than CFO, EdgeMix and high-resolution training improve local detail but still do not fully recover CFO, and P2 increases prediction resolution but also adds noisy detections. The remaining limitation therefore appears to be weak-class visual ambiguity rather than a single architectural bottleneck.

VI. DISCUSSION

The results support a consistent conclusion: the plain YOLOv11s baseline is a strong detector for DsPCBSD+, and most architectural additions do not improve it under the tested conditions. The baseline remains best overall, while attention modules, P2 detection, SAHI, EUCB, BiFPN, EdgeMix, and high-resolution training all remain below it.

This does not mean these methods are generally ineffective. Rather, adding complexity is not automatically beneficial when

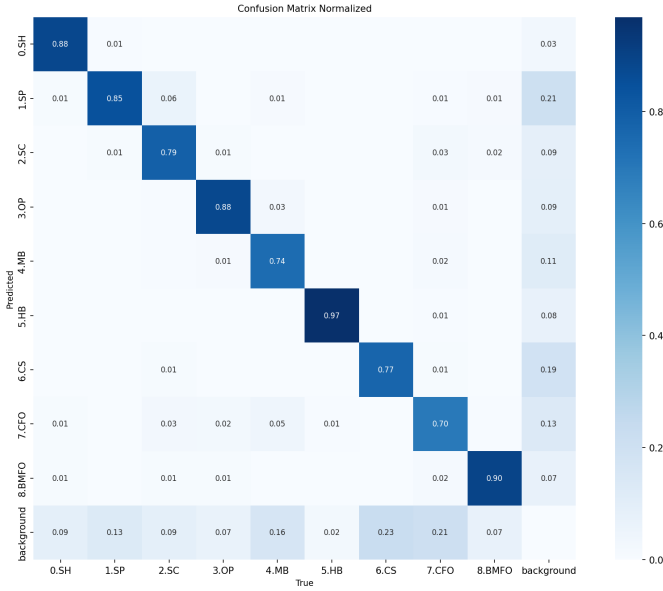


Fig. 4. Normalized confusion matrix of the best-performing model, the YOLOv11s baseline. Class indices follow Table I. Thin and irregular conductor defects, especially CS and CFO, account for much of the remaining confusion.

the starting model is already strong. In this dataset the weak classes appear to be limited by visual ambiguity, low contrast, and possible annotation difficulty: attention can reweight features but cannot create missing visual separation; P2 can increase prediction resolution but may also increase false positives; neck fusion can improve some thin features but does not fully solve class ambiguity. The most constructive signal comes from input- and neck-level changes—EdgeMix and high-resolution training stay close to the baseline, suggesting that preserving local texture and boundary detail matters, while BiFPN and EUCB remain competitive but insufficient on their own.

Deployment note. The baseline YOLOv11s is already lightweight for real-time inspection—9.4 M parameters and 21.5 GFLOPs at 640 px in its reference configuration [11]—and every module we evaluated only adds parameters and computation on top of this footprint. Having shown these additions yield no accuracy benefit, they are also unattractive on efficiency grounds; a precise per-model latency and FLOP

TABLE III
GROUND-TRUTH DEFECT BOX-SIZE STATISTICS ON DsPCBSD+, COMPUTED OVER 13,724 BOXES IN THE TRAINING SPLIT (SIZES AT THE 640 PX WORKING RESOLUTION).

Statistic	Value
Mean box (W×H), px	110 × 102
Median box max-side, px	79
Median box area (% of image)	1.02
Instances <32 px (COCO “small”)	8 %
Instances 32–96 px (mid)	48 %
Instances ≥96 px (large)	44 %

comparison on the target edge device is left to future work.

Limitations. The study is limited to one dataset and one backbone scale (YOLOv11s). Some later experiments were fine-tuned from a shared extended-baseline checkpoint rather than trained from scratch, which introduces a confound (Section IV-F); their results should be read as marginal tests under limited compute rather than universal conclusions. In addition, some inference strategies such as SAHI were tested only at inference time; training on sliced images may produce different behavior.

VII. CONCLUSION

This work evaluated YOLOv11s modifications for PCB surface defect detection using a stage-wise pipeline perspective, grouping experiments into input/data, backbone, neck, detection-head, and inference-level modifications. The plain YOLOv11s baseline achieved the best overall result ($\text{mAP}@0.5 = 0.839$, $\text{mAP}@0.5:0.95 = 0.499$), and none of the tested modifications surpassed it. The main practical finding is that the hardest PCB defects are not solved by simply adding attention, extra prediction heads, weighted fusion, upsampling modules, or sliced inference. Input- and neck-level changes were the most competitive, but weak classes such as conductor scratch and conductor foreign object remained difficult. Future work should focus on class-targeted data improvement, high-resolution training from scratch, and specialized weak-class detection pipelines rather than further architectural stacking.

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